

README.md — Adventure Works Sales Dashboard

Project Overview

This project is an interactive business intelligence dashboard built in Microsoft Power BI using the **Adventure Works** dataset.

The dataset was originally provided in Microsoft Excel format and transformed into an analytical model using Power Query.

The goal of the project was to analyze:

- sales performance
- profitability
- customer behavior
- product efficiency
- reseller performance
- country-level trends

Data Preparation Process

Steps completed:

1. Data Import

Imported raw data from Excel into Power BI.

2. Data Transformation

Used Power Query for:

- cleaning null values
- formatting columns
- adjusting data types
- creating calculated columns
- preparing clean tables for modeling

3. Data Modeling

Built a relational model using **Star Schema** approach:

Fact Table 'Sales_data':

Contains transactional data:

- Sales Amount
- Product Cost
- Unit Price
- Customer IDs
- Product IDs
- Reseller IDs
- Dates

Dimension Tables:

- Customer_data
- Product_data
- Sales Territory_data
- Reseller_data
- SalesOrder_data
- Date_data

Relationships were created using key columns between fact and dimension tables.

4. DAX Measures Created

- Profit Margin %
- Average Order Value (AVO)
- Median Order Value
- Customer Count
- Reseller Count
- Purchase Frequency
- Growth vs Previous Year

5. Parameters Created:

- for decomposition tree: Profit, Sales Amount, Total Costs, Total Orders
- Top N Reseller
- Top N Customers



Dashboard Pages & Business Value

1 Overview Page

Main indicators:

- Sales Amount = **51.9M**
- Profit = **5.8M**
- Profit Margin = **11%**
- Orders = **23.5K**
- Customers = **17.6K**

Departments:

- Executive Management
- Finance
- Sales Department

Business Questions Answered:

- Is the company growing year over year?
- Are profits increasing with revenue?
- How many customers place orders?
- What sales channels perform better?

Recommendations:

- ✓ Continue scaling high-performing years like FY2020
- ✓ Improve margin, not only revenue
- ✓ Expand stronger sales channels (Internet = 92%)
- ✓ Monitor reseller performance separately

2 Product Analysis Page

Visualizations:

- Top products by orders
- Top products by profit
- Orders and Profit Margin by category
- Subcategory sales
- Unit Price vs Standard Cost
- Drill through the products (price and cost)

Departments:

- Product Team
- Procurement
- Marketing
- Sales

Business Questions:

- Which products sell the most?
- Which products generate the most profit?
- Which categories underperform?
- Are pricing levels optimal?

Key Insights:

- Bikes generate highest sales
- Accessories have high volume
- Top profitable products: Mountain series bikes

Recommendations:

- ✓ Increase stock of top-selling bikes
- ✓ Bundle accessories with bike purchases
- ✓ Review low-performing SKUs
- ✓ Optimize pricing based on cost margins

3 Country Analysis Page

Visualizations:

- Profit by country
- Margin by country
- Orders by country
- Drill-down by product/category/channel

Departments:

- Regional Management
- Finance
- Strategy Team

Business Questions:

- Which countries generate highest profit?
- Where are margins weak?
- Which channels work best in each country?

Key Insights:

Top markets:

1. United States
2. Australia
3. United Kingdom

Recommendations:

- ✓ Prioritize US and Australia growth investment
- ✓ Rework low-margin countries

4 Customer Analysis Page

Visualizations:

- Customers by country
- Orders vs AVO and Median OV by country
- Top customers by order count
- Customers by city map

Departments:

- CRM
- Marketing
- Sales

Business Questions:

- Where are most customers located?
- Which countries have highest order values?
- Who are repeat customers?
- Which cities have strong demand?

Recommendations:

- ✓ Launch loyalty campaigns for top buyers
- ✓ Focus acquisition in strong cities
- ✓ Run location-based marketing campaigns tailored to specific countries and cities to improve conversion rates and optimize ad spend
- ✓ Upsell in countries with high AVO

5 Reseller Analysis Page

Visualizations:

- Profit margin by city
- Orders vs margin by business type
- Orders vs margin by country
- Reseller city map

Departments:

- B2B Sales
- Partnerships
- Distribution Team

Business Questions:

- Which reseller types are most profitable?
- Which cities have strong reseller networks?
- Which partners reduce margin?

Recommendations:

- ✓ Focus on profitable reseller types
- ✓ Renegotiate low-margin partnerships
- ✓ Expand in profitable reseller cities
- ✓ Segment reseller programs by type



Strategic Business Recommendations

1. Focus on the most profitable products

The company should focus not only on sales volume but primarily on product profitability. This means:

- increasing marketing support for high-margin products
- ensuring consistent stock availability of these products
- using them as core products in advertising campaigns

It is also advisable to review low-margin products:

- increase their price
- or optimize costs
- or gradually remove them from the ассортимент

2. Development of the online sales channel

Since most orders come through the online channel, the company should:

- invest in digital marketing
- improve website usability and the purchasing process
- implement personalized product recommendations

Additionally, the company can:

- launch online promotions
- use email marketing
- test new customer acquisition channels

3. Use of geo-targeted marketing

Since the data shows varying performance across countries and cities, the company should:

- run local advertising campaigns for different regions
- adapt offers to specific markets
- use different pricing or promotions depending on the region

For example:

- in strong markets (like the USA) — scale sales
- in weaker markets — stimulate demand through discounts

This will allow more efficient use of the marketing budget.

4. Customer retention efforts

Acquiring new customers is usually more expensive than retaining existing ones, so the company should:

- implement loyalty programs
- create personalized offers
- focus on repeat sales

5. Optimization of reseller partnerships

Since profitability depends on partners, the company should:

- identify the most profitable types of resellers
- review cooperation terms with less effective ones
- focus on developing the most beneficial partnerships (analyze the map to see where most customers are concentrated)



Tools Used

- Microsoft Excel
- Microsoft Power BI
- Power Query
- DAX
- Data Modeling



Final Result

Developed a full interactive BI solution that helps stakeholders monitor the business indicators, identify profitable products, optimize markets, and improve strategic decision-making.



Portfolio Value

This project demonstrates skills in:

- Data Cleaning
- Power Query ETL
- Star Schema Modeling
- DAX Calculations
- Dashboard Design
- Business Analysis
- Storytelling with Data